Abstract

As IMO MEPC.378(80) Guidelines transition toward structured enforcement convergence across Port State Control (PSC) regimes, biofouling management is shifting from voluntary guidance toward structured operational oversight. While in-water cleaning offers a cost-effective alternative to dry-docking, it requires stringent adherence to environmental standards and technical documentation across different European regions. Adriatica D.O.O. provides engineering management services to plan, coordinate, and certify in-water cleaning operations. Although Adriatica D.O.O. operates primarily in the Adriatic and Mediterranean regions, our compliance framework is aligned with broader European coastal regulatory expectations under the Paris Memorandum of Understanding (Paris MoU) framework [5, 9], ensuring continued compliance for 25m+ and 50m+ vessels without disruption to operational scheduling.

The European Regulatory Landscape & Operational Risk

Under the IMO MEPC.378(80) Guidelines [1], biofouling management is progressing from voluntary guidance toward structured enforcement convergence. European coastal regimes, including the Mediterranean, Adriatic, and Baltic Seas, are intensifying scrutiny of arriving vessels [5]. Regional enforcement intensity varies; for example, the Baltic Sea region is subject to heightened environmental sensitivity under HELCOM coordination frameworks [9]. For motor yachts and sailing vessels (25m+), reliance on reactive cleaning leads to:

- High costs of unscheduled dry-docking.
- Charter schedule disruptions and operational downtime.
- Regulatory non-compliance penalties across different jurisdictions.

In-Water Cleaning: A Strategic Alternative

In-water cleaning (cleaning without hauling the vessel) provides a sustainable solution when executed professionally. It offers:

- Cost Efficiency: Cost reductions typically ranging between 60–70% compared to unscheduled dry-docking [8].
- Operational Continuity: Cleaning performed during guest turnaround or downtime.
- Technical Focus: Specialized removal of macro-fouling from niche areas (propellers, thruster tunnels, rudder pintles).

The Adriatica Engineering Management Framework

Adriatica D.O.O. manages the entire cleaning process through a structured approach, ensuring quality and compliance. It is important to note that Adriatica D.O.O. does not perform physical cleaning operations; we design, manage, and technically certify the process.

- Pre-Cleaning Risk Assessment: Technical evaluation of fouling levels to determine the necessary cleaning intensity based on the vessel's operational history. Technical diagrams identifying all vessel-specific high-risk zones.
- Certified Partner Coordination: Management of certified diving teams equipped with required technologies (e.g., vacuum-based systems for debris capture) to meet stringent environmental standards in diverse ports like Monaco, Tivat, Dubrovnik, or Kiel.
- Environmental Control & Debris Capture Compliance: Ensuring diver technology selection adheres to debris containment and sensitivity regarding discharge in European coastal waters, particularly in ecologically sensitive areas like the Baltic or specific Mediterranean marine reserves. In jurisdictions such as those under HELCOM coordination or strict port authority enforcement (e.g., Monaco), failure to contain cleaning debris may trigger Port State Control or port authority enforcement action, including administrative penalties or temporary operational restrictions, depending on jurisdiction [3].
- Quality Assurance & Reporting: Oversight of the operation, including high-resolution photographic documentation, ensuring the process is logged in the vessel's Biofouling Record Book (BFRB) [6].

Case Study: Economic Comparison

Feature	Scenario A: Dry-Docking	Scenario B: Managed In-Water Cleaning
Cost	€30,000	€6,000
Downtime	4 Days	1 Day
Charter Risk	High (Cancellation Likely)	Negligible
Environmental Compliance	Indirect (Yard documentation)	Direct (BFRB + Photographic Evidence)

- Note: Representative mid-season comparison based on European operational averages.
- Conclusion: Managed in-water cleaning provided verified compliance with significant financial and operational advantages.

Conclusion & Call to Action

As 2026 approaches, proactive biofouling governance differentiates compliant operators and protects charter continuity in high-demand Mediterranean seasons. Adriatica D.O.O. invites yacht operators to integrate managed in-water cleaning into their compliance roadmap ahead of 2026 enforcement convergence trends. We position our clients ahead of enforcement curves through strategic management, controlled documentation, and proactive inspection oversight in Mediterranean & European coastal waters.

References & Regulatory Standards

[1] IMO MEPC.378(80): *2023 Guidelines for the Control and Management of Ships' Biofouling to Minimize the Transfer of Invasive Aquatic Species.*
[2] IMO CII Framework: *Technical guidance on the Carbon Intensity Indicator (CII) and its operational efficiency impacts.*
[3] BIMCO / Clean Shipping Alliance: *Industry standards for in-water cleaning and biofouling documentation, emphasizing strict debris capture in sensitive areas.*
[4] IMO PPR Sub-Committee Reports (PPR 13, 2026 Preparatory Work): *Detailing technical measures and enforcement pathways for biofouling management.*
[5] Paris Memorandum of Understanding (Paris MoU) Guidance Notes: *Port State Control procedures and inspection criteria for biofouling documentation and compliance.*
[6] IACS Unified Requirements and relevant classification society rules (RINA, DNV, Bureau Veritas): *Concerning Biofouling Management Plan documentation and compliance in alignment with regional environmental expectations.*
[7] Australian/New Zealand Biofouling Standards (BIOF125): *Mandatory biosecurity models serving as benchmarks for global regulatory alignment.*
[8] Adriatica D.O.O. Engineering Archives: *Internal case studies on superyacht operational efficiency in European & Mediterranean regions.*
[9] Helsinki Commission (HELCOM): *Environmental* guidelines for the Baltic Sea area regarding biofouling and debris containment.

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